# MSX and SPECTRAVIDEO

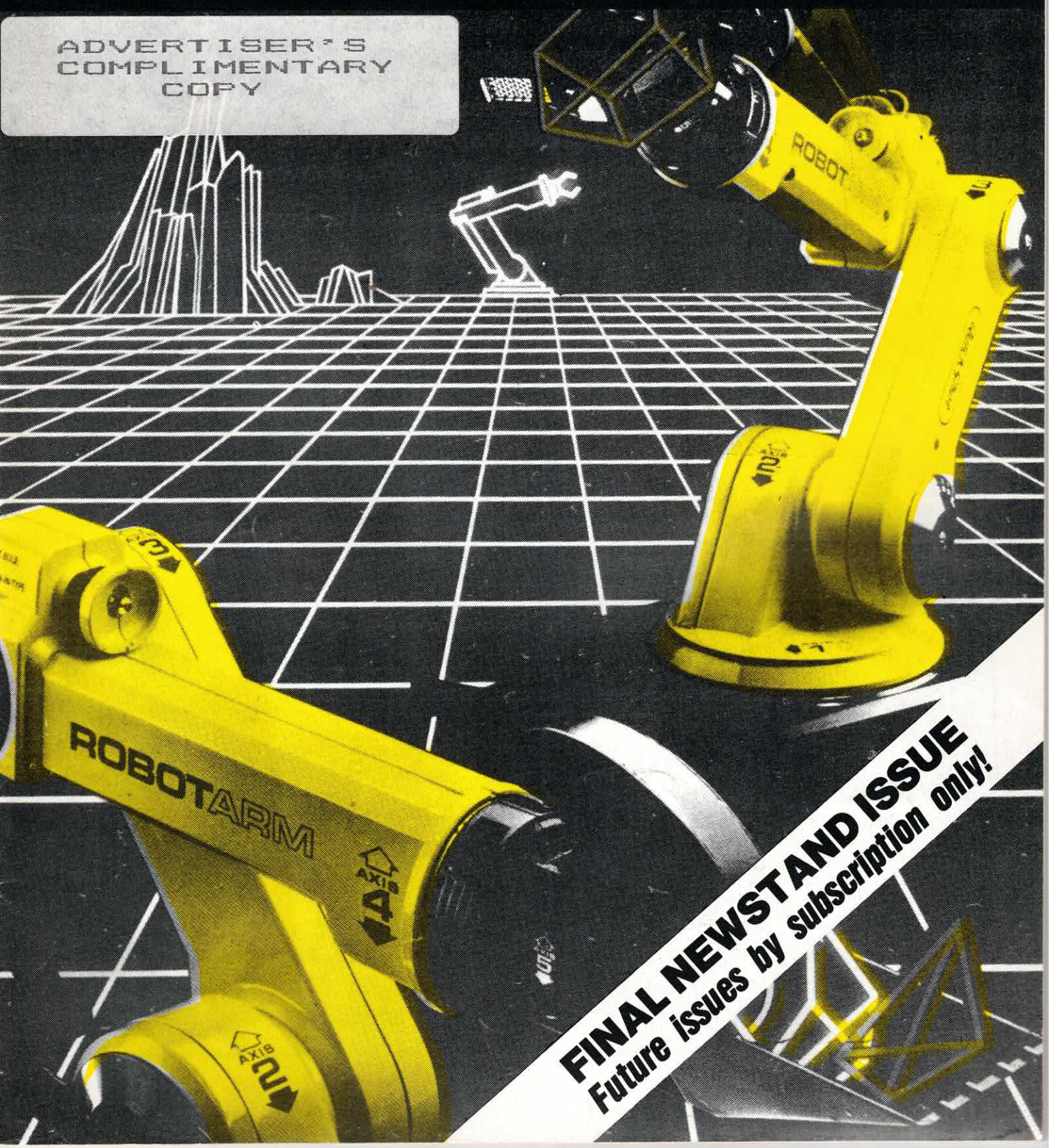
# COMPUTER FORUM

Volume 1 No. 6

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MSX and SPECTRAVIDEO
# COMPUTER FORUM
Magazine

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## EDITORIAL

*This is the last issue of Computer Forum available at newsagents. No, we are not ceasing publication. Computer Forum will continue to be available in its current format, but only by subscription.*

The reasons for this move are purely economic. We have not lost faith in the product or its subject matter but we must face facts. To produce sufficient quantities of a magazine for national release to newsagents places a great burden on the purse strings and has to be justified by paid advertisements. I think you know as well as I do how interested certain people are in advertising their involvement in our branch of computing.

This situation can and must change. However, in the meantime, we feel it is more important to provide continuing support to those people who have invested in these people's products. Please photocopy the back page and get your subscription in as soon as possible.

Happy computing ...

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# THE FUTURE FOR MSX AND SPECTRAVIDEO

*The Editor, Brod Brown, looks at some recent developments.*

A recent visit to Spectravideo in Hong Kong revealed some interesting news. The SV318/328 computers have not been in production for some time. The SV728 and SV738 MSX computers are no longer being built except on special order. MSX and SV software are no longer available from that company. Plans for MSX2 have been shelved for the time being. Spectravideo have a range of IBM compatibles and one of these, the X'press 16 can accept the SVI-811 MSX game cartridge adaptor.

On the home front, telephone conversations with MSX hardware and software distributors were pretty negative. The major software houses advised that MSX software was no longer being distributed *'until MSX computers start moving again'!* The major hardware distributors were not much more enthusiastic.

Mitsubishi, who, through AWA-Thorn also distribute the Amstrad, have a bundle of MSX computers in their warehouse and are looking at several ways of *disposing* of them. A spokesman for that company said that *'the Amstrad and the MSX systems were given equal exposure and the MSX machine could not compete'*. Must have been a lot of ads that I missed! Sony were also critical of the response to MSX in Australia. Once again, as a consumer, I have been told all about their TV and stereo products but not MSX. They have an MSX2 computer in this country, but it appears to be only for evaluation purposes — we can't get one, so we cannot comment further. The agents for JVC advised that there was no intention to bring in their MSX machines in the forseeable future. National Panasonic are concentrating on their business range and Pioneer will not be releasing their PX7 due to a high landed cost. At least this was a more believable reason for not doing anything.

A brighter note was sounded by Toshiba who advised that *'we hope to release our MSX2 system after Christmas'*. Sanyo are also starting to move in ever increasing circles to appoint *suitable* dealers for their MSX range.

## LACK OF FUNDS?

At this point it must be mentioned that this information was provided in response to our requests for advertising and information. The problem facing these major hardware houses, appears to be the fact that their parent companies, which reside in a long skinny land to the north of us, have virtually only supplied them with stock and a very small budget for promotional purposes. When you compare the amount of promotion that Commodore and Apple did (and are still doing), that budget must have been almost non-existant.

While, as a business organisation, we can appreciate that sales figures and profit levels must be used as guidelines a large percentage of product success is attributed to marketing attitude and clever (not shifty or sneaky) promotion.

Personally, I have never been impressed by the Commodore 64, with its cumbersome sprite system, difficult language and slow disk drives but I take my hat off to their marketing attack and attitude. The next best thing to an advertisement in COMPUTER FORUM is satisfied customers spreading the word. How many people do you know who have C64 computers and are continually telling people about the latest this, that or the other which is available for it? Albeit that these products are required to bring the computer up to anywhere near the power of MSX.

## POSSIBILITIES

So, what is the answer? One problem, as I see it, is that the public were made aware of MSX2 shortly after MSX1 machines went on sale. Maybe they thought that by waiting, they could get into MSX2 for the same price as MSX1. Although MSX2 prices have not been disclosed, the specifications would seem to require a higher selling price than the current machines. The other failure lies in the fact that MSX computers have been sold by brand name. MSX is a *'standard'* and must be promoted as such. The MSX logo needs more exposure — IBM clones are sold purely on price and the fact of being IBM compatible. Can you imagine the effect of not being able to use IBM in the term IBM compatible.

The major problem with MSX is that the manufacturers are not only producing computers. They are already selling a range of appliances from which they are already making profits. To channel existing promotion funds away from those products would not make sense — we are told. If they are making x+ profits from these lines, why risk making x– profits on MSX. I wonder what they did when they were landed with microwave ovens, a few years ago? *How did the public know about them?*

There would seem to be two possible ways that MSX could be marketed in this country. One is to price the computers into the $1000 range and only have them available from specialised dealers who have the full range of software to back them. Another way is to keep them price competitive and have them stocked by everyone —KMART, WOOLWORTHS, toyshops etc who also stock a range of popular software titles. This would not be backed by any computer knowledge — you buy them in the same way as buying any other appliance, you only go back for warranty repairs. All other information about the computer is gained from magazines, books and user groups. Let's face it, you don't go back to the shop and ask how to cook a certain meal in your new microwave — you buy a book or go to demonstrations. I know of one computer distributor that expects its computer dealers to operate on the same percentage margin as those selling its range of TV sets and stereo equipment.

## DEALER OR DISTRIBUTOR

We are continually receiving letters from readers who no longer have a dealer in their area, or can't get certain items for their computers. Dealers cannot be expected to bear the whole brunt of promoting a product. Why spend large amounts of money advertising and hours of time demonstrating a computer which is not being promoted by the people whose brand name it carries. Far better to become

a dealer for a computer which virtually sells itself because of publicity, satisfied customers etc.

No, the problem definitely lies with the distributors and they cannot keep passing the buck to dealers and software houses — although the latter are not entirely blameless! Rest assured that Computer Forum will continue trying to convince these people to get off their b.... They are all on our mailing list and receive, and will continue to receive, Computer Forum as long as WE can afford to do so because we know they are reading it. *Where else can they find out about MSX?*

## THE FUTURE

MSX hardware might become a problem, although the beauty of MSX is that you are not stuck with just one manufacturer. Compare this with, say, Amstrad owners who, if they couldn't get an Amstrad disk drive would have to look at importing it themselves.

Spectravideo hardware is another matter. This is already in short supply, but we know of two Australian companies that are negotiating to make some peripherals for these computers.

Software for both types of computer will continue to be available as long as there is a market for it. If the major software houses are not handling it, then some dealers will be bringing it in themselves.

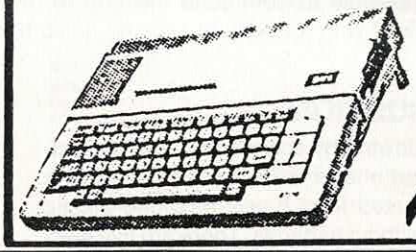
So don't lose faith. Computer Forum will continue to provide information for both types of computer. Armed with this you can start writing your own software as well. The main attraction of any piece of software is challenge and I can think of no better challenge than writing your own programs.

**END**


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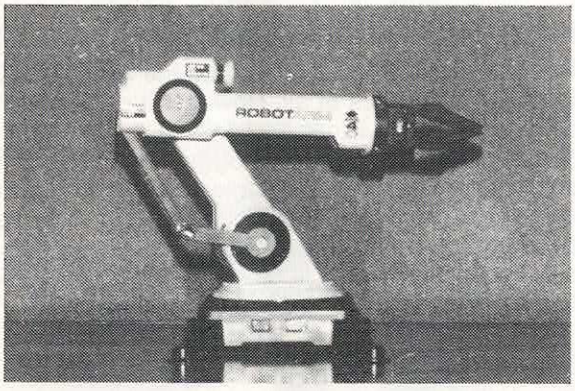
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# SPECTRAVIDEO'S ROBOTARM



*For the past few days we have been 'playing around' with a couple of Robotarms which we had flown in from Spectravideo in Hong Kong.*

The products we received were the SVI-2000 Robotarm with two joysticks and the SVI-2000B Robotarm with MSX Rogo language card. They can both be used with joysticks alone but you need the Rogo card to have an MSX computer control them.

First impression is that they are toys — being moulded in black and yellow plastic. This impression soon disappears when you start using the arm. The Robotarm will work with any standard joystick and this was the one we used first.

Four D size batteries power the Robotarm and after a mad dash to the shops to purchase these, we plugged in the joysticks and were up and running. The left joystick controls rotation of the base and forward and back movement of the lower arm with the trigger closing the 'hand'. The right joystick controls the vertical movement of the forearm and rotation of the 'wrist' with the trigger opening the hand. It took a while go get the co-ordination required for smooth control and we had a lot of fun as each of us tried our hand.

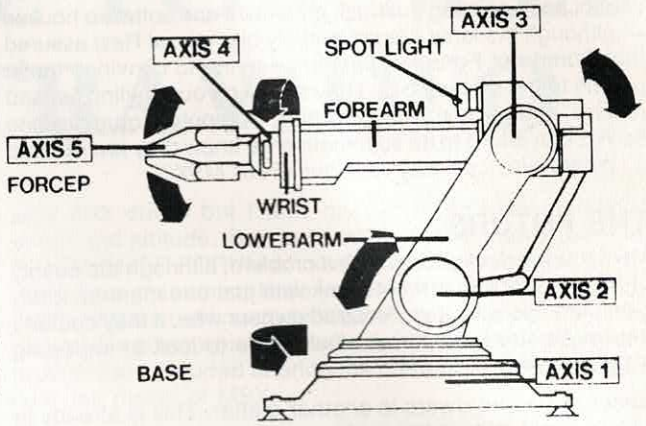
## ROGO LANGUAGE CARD

The SV2000B pack comes with the Robotarm, Rogo language card and two cables for connecting the card to the arm. With these connections made, the card was plugged into the cartridge port of our MSX computer and the computer switched on. Up came the message "WELCOME TO SVI-ROGO" followed by a ? prompt. What do we do now? Check the 150 page instruction manual that comes with the card.

Actually, the Rogo card is not just an interface for the Robotarm, but a special implementation of a Logo type programming language which can be used for other purposes as well. The main difference to Logo is that the turtle graphics have been replaced with some commands to control the Robotarm.

After reading the manual for about ten minutes, we typed in some commands and soon had the robotarm going through its paces in response to the 'program' we had just written. These programs or series of instructions can be saved to and loaded from tape — alas, disks are not catered for.

Rogo contains some Macro commands and one of these, SHOWARM, displays the arm on screen and have a graphic simulation of the results of your program. We found this handy to pre-test our program, as it was possible to command the arm to move further than its limits allowed with a resultant clicking noise from the motor concerned.




*At time of going to press, price and availability were still not known. However, interested readers should write to us for further details.*

## STURDY CONSTRUCTION

Our usual method of 'reviewing by screwdriver' got the better of us and we soon had one of the arms in pieces. The whole unit is well constructed from heavy moulded plastic and weighs over one kilo without batteries. There are five electric motors with heavy duty plastic gearboxes that rotate the base, move the lower arm, move the upper arm, rotate the wrist and open and close the hand. Whenever the hand is opened or closed, a little spotlight comes on to light the area where the hand is.

We were a little worried about the grip of the hand where 'littlies' are concerned. Although the hand grips firmly with rubber pads, it has a slip-clutch mechanism to stop excess pressure. "What a pity!" say some ankle-biter haters.

Both units came in beautifully packaged gift boxes and would make ideal presents for Christmas. The only bad point is that SV318/328 owners can only use the arm with the joysticks as the Rogo card is MSX only. However, we have an idea for our own card which will enable non-MSX computers to interface with the arm. More details next month.

At present, plans are underway to import some of these arms and Computer Forum is negotiating with the importers for a special deal for subscribers. At this stage, no prices can be given as fluctuations in the exchange rate make pricing difficult.

**END**

# MSX1 & MSX2 QUESTIONS ANSWERED

*by Ross Brisbane*

*This article is in response to many questions being asked by MSX1 owners about MSX2. Answers are based on using an MSX2 Sony Hit Bit.*

### Do all MSX2 peripherals run on MSX1?

No. Suppose you have a Video or Photo Digitizer. These peripherals won't run because they make calls to the more powerful Graphic Chip or Video Display Processor. However Disk Drives, Printers, Printer Plotters, Mouse, Joysticks, Graphic Tablets and Lightpens are MSX1 provided software to drive them does not make calls to the more powerful MSX2 display chip.

### Is Disk Basic or MSXDOS more powerful on MSX2?

No. I have tested MSXDOS from the Sony MSX2 and found no difference at all. However it does have to be pointed out that 3.5 inch Disk Drives in MSX2 are 720K capacity and the format program under Disk Basic prompts for Double — or Single-sided format. Sony are releasing a new floppy interface for MSX1 and designed for double-sided drives. To my knowledge, no commercial MSX software is being produced on 5.25 inch format diskettes.

### Do all MSX1 programs run on MSX2?

Yes. I have carried out extensive tests on MSX2 and have found 100% compatibility.

### Is Disk-based software coming out in MSX2 only?

No. A late listing from Sony indicates a great deal of software has been produced to run on both MSX1 and MSX2. The software is usually produced single-sided in commercial form.

### What companies will be marketing MSX2 in the near future?

This is not easy to answer at this stage. There are major negotiations going on right now and we should see some outcome from this very shortly.

### Will I have to trade my MSX in or sell it off?

No. Honestly, this is a misunderstanding of the whole MSX concept. Standardization is the safety buffer you have on your side. Your major cost is the software and hardware you run and plug into your system, NOT the CPU/Keyboard. However there are rumours of a possible MSX2 adaptor coming out.

### Can MSX ever become popular in Australia?

YES IT CAN! MSX is still a very powerful concept for the home/business market. To see a demonstration of MSX2 is to leave you quite astounded. I am looking at a mass market MSX2 at the moment, possibly selling for $699-$799. Considering its power, 128K RAM and 128K VRAM for graphics, this will place into the market a product with more power for your dollar than any other microcomputer ever created. Software is there in place already (MSX1). Over six hundred titles in fact!

*Ross Brisbane works for the Tomorrowland organization — a chain of eleven stores from Melbourne to Darwin marketing the Cleveland PC/XT IBM compatible.*

**END**


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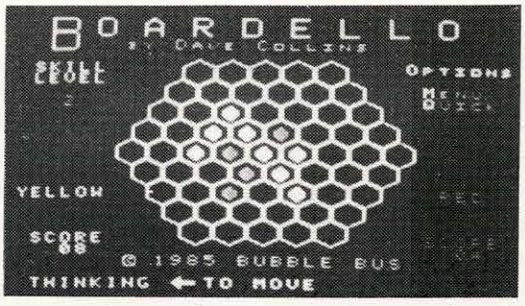
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If the order is under $25.00 please include $2.00 for postage & handling


# *Software NEWS & REVIEWS*




## BOARDELLO (MSX Tape)
### by Bubble Bus



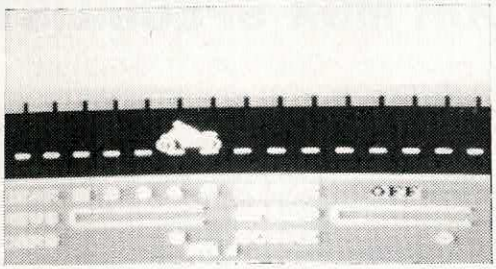

A board game for two players or one player against the computer with five different skill levels and a variable size board. In essence, this is a variation on the old board game of Othello — the object being to outflank your opponent and claim any of his pieces which lie between the piece you are playing and another of your pieces.

The one with the most pieces on the board, when no more moves can be made, is the winner. If you are stuck for a move, pressing the H key will make the computer move for you. Pressing the Q key while the computer is "thinking" will cause it to play the best move it has "thought of" so far.

Price: $19.95

*(Review copy supplied by Peter J. Fox Computers)*

## JUMP CHALLENGE (MSX Tape)
**by Martech**



This is the Eddie Kidd Jump Challenge. The aim is to jump as far as you can without losing control of the bike on landing. Before you can try your hand at jumping a motor cycle over cars, you have to prove your worth jumping a bicycle over oil barrels.

The procedure is to ride away from the ramp as far as you think is necessary for a good run up. Stop the bike and turn to face the ramp. Accelerate towards the ramp. The score you receive for a successful jump depends on where your back wheel hits the landing ramp.

The hardest part of this game is knowing how far down the approach road you have gone before turning around for your run up — no indication is provided.

The graphics are fairly simple except, without being morbid, the part where Eddie crashes. Anyway, it doesn't matter how bad the crash, he always gets up and waves to the crowd. Good use has been made of the illusion of speed, with the foregound moving faster than the background and sky. Better use could have been made of sound in this program.

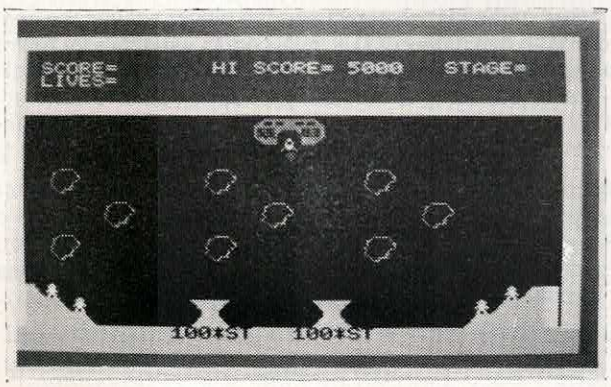
Price: $19.95

*(Review copy supplied by Peter J. Fox Computers)*

## FDS SOFTWARE (MSX & SV) Cassette SVPAK11/MSXPAK11

This pack has three titles on the tape — LUNAR RESCUE, SKY BASE, QUASIMODO.

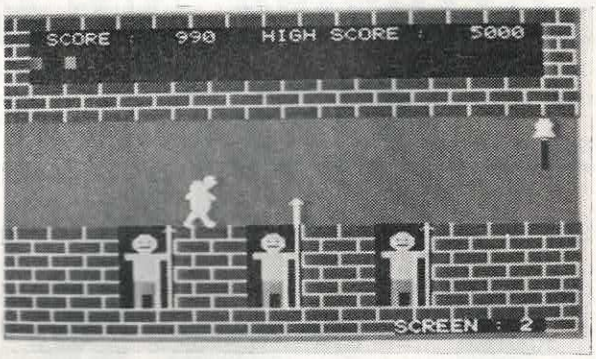
**Lunar Rescue** — You control the decent of the rescue module from the mother ship. You must avoid the asteroids on the way down to rescue the stranded astronauts one at a time. Once you have one on board, you can blast off for the mother ship while destroying or avoiding the asteroids. The more screens you successfully complete, the faster the asteroids move.




**Sky Base** — Your space fighter is on screen with four alien craft which you must destroy before they line you up and dive on you. After you destroy two screens of these craft, you can dock with your star ship and attempt to destroy the alien base which is heavily defended.




**Quasimodo** — The hunchback, Quasimodo, must make it across the screen to ring the bell. One screen has guards in pits which he must jump. In the next screen, the guards are armed with spears which are thrust at Quasimodo as he tries to jump. Other screens have spears coming down from the roof and flying objects which make it more difficult for him to jump the pits and ring the bell.




Publisher: Forrest Data Services
Recommended price: $16.00

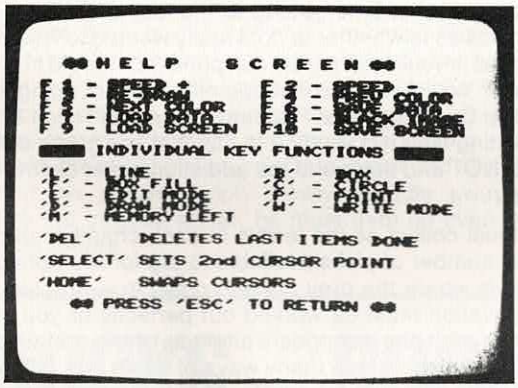
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# *Software NEWS & REVIEWS*

## VIDEO GRAFFITI — (SV & MSX TAPE) by Electric Adventures

This is an art program which allows you to draw and paint directly onto the screen. Video Graffiti is the best I have seen so far and great use has been made of pull down menus in the Apple Mac style.

Placing lines, boxes, circles and text on the screen is a dream with this program. The ability to save and load the screen via cassette is very useful and disks are catered for in the MSX version.





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Four pen sizes can be selected and there is a 'Spray Can' feature which can produce textured effects with up to 225 colours. Other nice features include being able to grab an area of the screen to be displayed elsewhere and the ability to dump the screen image to a printer in normal or enlarged image.

Distributed by Forrest Data Services, Video Graffiti will be available soon at your local dealer.

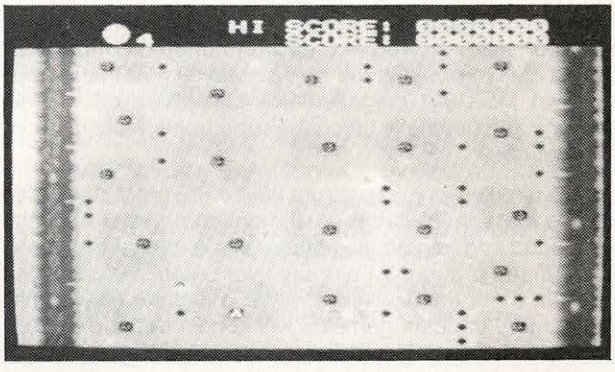
Price $19.95 (MSX disk $24.95)

## SENIOR CITIZENS USE SPECTRAVIDEO FOR BINGO

Taking advantage of a recently low-priced special from a local store, Rockingham Senior Citizens Club recently purchased a SVI318 Spectravideo to enhance their weekly game of Bingo. It was found that patrons being able to see the numbers called, as well as hearing them, was a big improvement.

Unable to find a suitable program on the market, one of their members, well conversed in other computers, was asked to produce a program. This program is now available free to non-profit clubs and charity organisations from the author Doug Stanborough (095) 28 1791 on production of a tape to put it on.

## LEUCOCYTE (SV AND MSX)




You control the warriors of the human body in this novel approach to a Pacman-style game. The body is under attack from bacteria and viruses and you must move out of the bloodstream to devour them before they invade the cells of the body or completely take over.

If you can collect an infection fighting nucleus from the bloodstream then this can be fired at any of the invaders of the cells before they overcome you.

Available on tape or disk, this machine code game has very good graphics, smooth movement and good sound effects. Our copy was supplied by Panda Soft International.

**END**

# Our first READERS' REVIEW

I'm sure many readers, like me, are tired of buying professionally written home entertainment software only to be disappointed. A short demonstration in the shop doesn't always expose a bad program's faults or, for that matter, the intricacies of a brilliant one.

Not all dealers have an in-depth knowledge of programs they sell and some are loathe to demonstrate software. So we rely on borrowing from friends. "Hey — if I can borrow it, why buy one?" No — if it's good enough I want my own original. I have very strong feelings about pirating as I hope all true enthusiasts have ... IF IT IS GOOD ENOUGH TO COPY THEN, BE FAIR, IT'S GOOD ENOUGH TO BUY. Think of it as an investment in the future. If you are tempted to pirate programs you are not only BREAKING THE LAW but doing untold harm to the future of our hobby because —
(a) The quality of programs available will go down.
(b) The cost of good programs will rise (someone has to buy it for you to copy).
(c) Small but promising software houses can be put out of business through pirating. Who knows what possible brilliance has been lost to us forever.

How do we know a program is worth investing in? One answer is to hire it first. Although MSX hire clubs are rare, a few are springing up and those run by retail outlets will usually deduct the hire cost from the price if you decide to buy. If, like me, you live miles from a software library, they will often mail out programs — but this cuts down the time you have it and top programs (usually the most expensive) can take a lot of use to reveal their true potential.

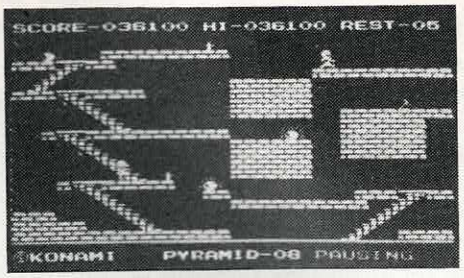
The only other answer I can think of is recommendation or non-recommendation by users. But if you don't happen to know many people with home computers or live in a remote area then, like me, you end up with hundreds of programs that I'd hate to work out the cost of. Maybe now that we have our own MSX/SVI magazine we can share each others opinions and hopefully gleen the truth behind all that advertising blurb.

Here is a review of my favourite adventure program. I intend to write more in the future, commenting on what I consider to be good and bad, to give others some idea of the content of the many programs I already own before they decide to invest in them themselves. I will also include playing hints on the more intricate games. I will concentrate on MSX format as there are so few SVI programs available that I'm sure we all know the good and the bad.

## KING'S VALLEY (MSX Cartridge)
### by Konami

The instant you start the game you will be amazed by the superb music which leaves you in no doubt that the adventure is set in the Egyptian pyramids. You control Vick "the notorious adventurer from Manchester" as he takes an archaeological adventure trip to the King's Valley (The Ancient Egyptian Valley of the Pharaohs) in search of the hidden mystery jewels in the Royal Tombs while avoiding the deadly Mummies that haunt the chambers.

Along with superb animation and beautiful use of the PSG for background music and sound effects, which I have come to expect from Konami, this game finds that very fine line between being too easy to hold your interest and too hard to keep it with absolute precision. Having owned this cartridge for over a year now I still get hours of regular enjoyment from it and can honestly say no other game has held my interest so consistently for so long and I know that I'm not alone in my view. Not only does it require accurate joystick control, you have to use the old grey matter a great deal to succeed but, without the harassment of a time limit, it is very satisfying.




The adventure starts by giving you a map of 15 chambers with a goal at the end. You don't know what is at the goal until you get there which can take weeks of practice. After spending all that time getting to the last chamber I was in two minds as to whether or not I really wanted to know. Once I reached it would all interest disappear? ... Would I be disappointed? Well, I was not disappointed! I'm not going to spoil your fun by telling what happens when you reach the goal (frustrating isn't it?). Suffice to say that the game definitely DOES NOT end there and the addiction never seems to end either.

Vick must collect all the jewels in each chamber using the limited number of picks available to dig for the buried ones — this is where the grey matter comes in — the sequence of excavation must be worked out perfectly or you run out of picks (each one disappears after use) the fascinating thing is that there are many, many ways of doing this. Meanwhile the deadly Mummies have to be avoided or temporarily destroyed with the strategically placed daggers, remember — Mummies are dead already so they only disappear for a short time before getting over the shock and coming back to haunt Vick. On collecting all of the jewels the doors to the next chamber appear. You have to open them and walk through to the next scene. Many of the chambers take up two screens and the sideways scrolling as Vick walks off to the left or right is beautifully done. The layout is neat and uncluttered, the clarity of which is a great help when working out the excavation sequence. The animation is delightful, and the action of the dagger thrown by Vick, which spins through the air and bounces off walls etc. is very realistic and spectacularly done.

Konami's attention to detail both graphically and musically coupled with their excellent storylines must make them one of the best producers of home entertainment programs in the world — THE best in my opinion. If you want to make a good investment in many hours of satisfying entertainment, then you can't go wrong by purchasing King's Valley. Living in Australasia we have to pay dearly for our games programs, especially in convenient cartridge form (Aussies count your blessings — the Kiwis pay a lot more than you do!), you won't regret a cent of the price in this case.

## PLAYING TIPS

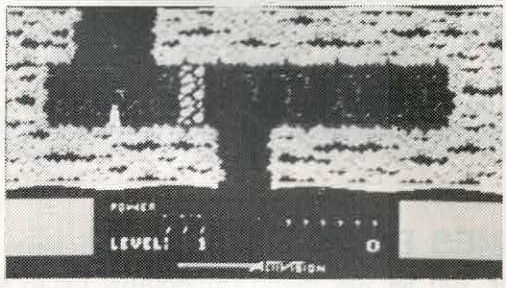
1. When making Vick jump — you will be surprised how far he can jump from a ledge if you press the trigger just as he appears to have one foot in mid-air.
2. When digging for a jewel, either leave ledges to jump out onto or take a pick in with you to dig out.
3. Use the most distant picks first — it gets easier then and you don't risk digging a hole between the jewel and the remaining picks (you can't jump with a pick in your hand).

4. Study the movement patterns of each colour Mummy as it can help you get around them.
5. Draw the Mummies to another part of the chamber, pop them off and dash back to complete excavations.
6. Dig holes for a Mummy to fall through. This can leave you unhassled while collecting jewels and provide a quick means of escape.
7. Vick walking over certain places triggers a wall to be built. Take note of these — it can help to keep Mummies at bay. Be careful not to wall up a pick or, worst of all, Vick himself.
8. Place the daggers thoughtfully where you may need them later. This can save your life.
9. When you have finished a chamber, stand beside the open doors and pop off as many Mummies as you can before leaving and add to your score. For each 20,000 points you get another life.

*Thanks to Patricia Perris of Auckland, New Zealand for this article.*

## H.E.R.O. (MSX Tape)
### by Activision

In this game, you play the part of R. Hero who has four lives to start with. His mission is to rescue the miners trapped in the mine shafts below. His equipment consists of a helicopter backpack, a laser beam and a number of sticks of dynamite. He must traverse the shafts and corridors blasting away the rubble with the sticks of dynamite or the laser and all the while watching his power supply slowly dwindle away. After reaching the trapped miner, he must then go even deeper to rescue others.




Interest is held in this game by increasing the difficulty of getting to the trapped miners and introducing various hazards. One particularly good feature is the use of lights in the shafts. If you bump into these, you are then left in almost total darkness. The various hazards encountered are hot lava rocks and rivers, octopi, spiders, butterflies, dead ends and moving walls.

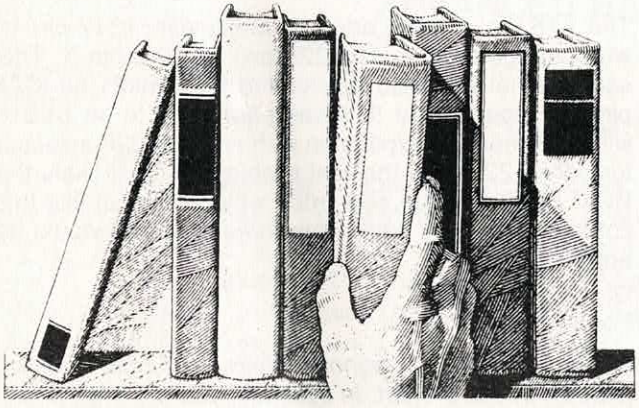
This is a very engrossing and entertaining game. Graphics and sound effects are good. Its appeal lies in the simplicity of the plot, anyone can play it, and the different challenges encountered on the way. Personally, I believe this to be one of the better games around for MSX machines.

RATINGS: Graphics — 8 out of 10
Sound — 8 out of 10
Overall — 8 out of 10

*Thanks to John Knight of Cranbourne, Victoria for this review.*

**END**

# BOOKS • BOOKS • BOOKS • BOOKS



## THE MAGIC OF SPECTRAVIDEO
### by Bernard L. Burke

Tape based Editor Assembler for SV318/328.

This package is really a tape assembler with book of instructions and reference manual. The first two chapters deal with number systems and the general way that information is stored and handled by the computer.

The Memory Map is discussed in chapter 3 which also has a bank switching routine for accessing the unused RAM area in a SV328. The next chapters cover the area of RAM used to store a program and useful addresses which control the storage of variables and arrays, top and bottom of memory pointers and the stack area.

Chapter 8 contains more useful addresses in the SYSTEM area together with a routine to autorun a program which has been CSAVE'd. The next six chapters discuss the Video Display Processor and the Video Ram. Several routines are included to demonstrate manipulation of the screen.

The rest of the book covers a machine code and how to use the Assembler which is provided on the enclosed tape. There are more routines and programs in this section for demonstration of how machine code can be used from Basic to enhance the performance of graphics and the good part is that all of these are also on the tape for those who do not wish to type them in.

In the appendix, the book provides the Z80 machine code mnemonics (symbols), locating ROM routines, I/O port table, a list of some ROM routines, directory of the programs on tape, basic statement handler, hooks and reading input devices.

The tape based assembler is a little unusual to use at first, but after entering the examples given and experimenting with a few of our own routines that we knew to work, things became more familiar.

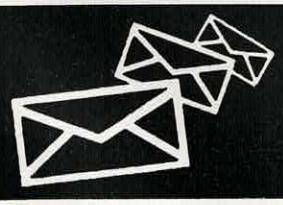
This package has been around for some time and, being a South African product, availability is limited.

Review copy supplied by TJ Soft.

Publisher: Intersoft
160 pages

**END**

# MAILBOX



## GAMES BOOK HELP

Thank you for the report on Andrew Lacey's MSX Games Book (Vol. 1 No. 4). I have been trying to type in many of the programs on my SV738 X'press but have failed completely with the programs requiring the machine code support. Now you have come to the rescue and I hope that you will be able to supply the necessary information to enable me and others who have purchased the book to make good use of it. Thank you for producing an excellent magazine and my best wishes for its success.

A. F. Evans
Whitianga, N.Z.

*We have been advised (but unable to confirm) that an errata sheet has been produced for this book. At present, we are still looking at modifications to the support program for use with disks — it works fine with tape.*

## MSX2 INSIDE INFORMATION

Rumour has it that Sony intends releasing their MSX2 later on this year ... it is also rumoured that it will come with TWO 3.5'' drives built in and expected to retail for around $1500. If these rumours eventuate, then MSX2 will be just the thing to cut Commodore 64/128 sales to the bone, and also anything else in the same price range which is not called Amiga or MSX2.

Due to the 100% compatibility (upward MSX/MSX2), I hope that the current software shortage will rectify itself within a few years after release of MSX2, as more people jump onto the MSX bandwagon here in Australia. The current situation is absolutely dreadful, with COMPUSHACK discontinuing all sales of MSX software.

Apparently Toshiba supporters may have to wait for the Toshiba MSX2 as there seems no sign of it being released. Either Toshiba are keeping quiet or they have released an MSX2 that I have missed. I hope the above information proves to be useful, despite it being mostly rumours.

As for upgrading a Laser MSX to MSX2 with 'an optional adaptor' — I'll believe it when I see it! As far as I know, MSX2 will use a different chip set, whilst retaining MSX compatibility. The graphics chip will be the TI V9938, and some other peripheral chips around the Z80. If the Laser MSX can be upgraded, then they've done well, and I can't wait to get my hands on one!

Roger Stockburger
Richmond, N.S.W.

*Thanks for the information, Roger, and it does appear that Sony have a couple of HB-G900P Genlocker MSX2 machines in Australia — one in Queensland, one somewhere else, but none in our office — yet! Toshiba advise of their MSX 2 probably being released in the New Year. Pricing of both machines is going to be a problem — with the Aussie dollar having found its own bottom of the harbour scheme. MSX2 machines in Japan range from 85,000 to 200,000 Yen (check your newspaper for latest exchange rates).*

*Latest news on the Laser is not good — with 'samples being available late 1986, early 1987'. The feasibility of an MSX2 adaptor has been denied by many people but consider this —*

*''I/O port 88-8B VDP (9938) for MSX-1 adaptor. These addresses are for the adaptor that includes 9938 video chip and provides MSX2 features for MSX1 system.''*

*Our source — MSX2 Technical Data Book published by ASCII Microsoft. Of course, this only means that allowance has been made for an adaptor but hopefully someone will make one.*

## MEMORY RE-MAPPING

Having upgraded my SV318 to 64K I wanted to re-map the lower 32K in bank 21 to become bank 22 for non-CP/M purposes. Unfortunately Basic refused to boot up in this configuration.

The A15 input of the address multiplexer IC17 pin 13 was changed to be Bank22 (low) at IC46 pin 3. This same signal was also substituted for Bank21 on IC24 pin 9. I thought that this was equivalent to an SV318 with 16K memory expansion with an extra 32K selected for banks 22. I feel the real problem is most likely the ROM Basic boot up sequence which may not like this configuration. Any help, comments or advice would be appreciated.

G. Thornton
Forest Hill, Victoria

*It is very difficult to diagnose where your logic is wrong (if at all) in this project. In essence, you are endeavouring to emulate an SV807 64K RAM cartridge. This cartridge will allow you to switch in two 32K RAM pages into either of banks 02, 22 or 32 without affecting the ROM in bank 01.*

*Looking at the circuit diagram for the SV807, our immediate thought was that you need to take care of the RAMDIS (low) signal which, on the SV807, is pulled low by any combination of switches selecting banks 02, 22 or 32. To illustrate, if a low BANK 22 is output by the PSG, and the switch for selecting BK22 is closed (grounded) these two low signals are ORed to produce a low on the RAMDIS line going back to the computer. I have not been able to test your alterations, but hope that this small piece of information is of assistance.*

*Continued on page 23.*

COMPUTER FORUM Magazine
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# BASIC TUTORIAL — Part 6

## SAVING THE SCREEN IN MSX BASIC

Last month we mentioned about saving the Spectravideo VRAM to tape or disk using the CSAVE with the ",s" option. The same function can be performed with MSX DISK Basic, except that the BSAVE and BLOAD commands must be used.

Note that this will NOT work with cassette!

Write a program to draw some graphics on the screen and make the last line

```
BSAVE"SCREEN",0,16383,S
```

When the program is run, the entire contents of the VRAM will be saved to the file SCREEN. Now type NEW and enter SCREEN2:BLOAD"SCREEN",S
and the original screen image should appear.

## ADVANCED BASIC COMMANDS

### PEEKING AND POKING

These are two commands which seem to amuse people when they first encounter them. There is nothing funny about them - they are very serious and if used incorrectly, can have disastrous results. The PEEK command can be used quite safely even with the ROM area from memory locations 0 - 32767. Use the POKE command with caution because it can alter some important areas of memory and the only solution is to switch off the computer to get out of trouble.

### The PEEK command

This command is used to examine or PEEK at a memory location. It takes the form - PEEK(memory location) where memory location can be given as a value between 0 and 65535 and must be used within brackets - for example PEEK(33000) or it can be a numeric variable as in PEEK(I)

A good example of how PEEK can be used, is to examine the area of the System RAM where the keyboard matrix is located and check for a key being pressed. This can sometimes be used in place of the INKEY$ command when you want to

wait for a certain key to be pressed. The keyboard matrix is as follows -

SV318/328 KEYBOARD MATRIX

| MEMORY LOCATION | | VALUE CONTAINED IN MEMORY LOCATION | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 254 | 253 | 251 | 247 | 239 | 223 | 191 | 127 |
| M | 64885 | ) | ! | @ | # | $ | % | ^ | & |
| E | " | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| M | 64886 | * | ( | ; | " | < | + | > | ? |
| O | " | 9 | 9 | : | ' | , | = | . | / |
| R | 64887 | | | | | | | | |
| Y | " | | A | B | C | D | E | F | G |
| | 64888 | H | I | J | K | L | M | N | O |
| L | 64889 | P | Q | R | S | T | U | V | W |
| O | 64890 | | | | { | ~ | } | BACK | UP |
| C | " | X | Y | Z | [ | \ | ] | SPACE | ARROW |
| A | 64891 | | | LEFT | RIGHT | | | | LEFT |
| T | " | SHIFT | CTRL | GRPH | GRPH | ESC | STOP | ENTER | ARROW |
| I | 64892 | F6 | F7 | F8 | F9 | F10 | | | DOWN |
| O | " | F1 | F2 | F3 | F4 | F5 | CLS | INS | ARROW |
| N | 64893 | | | | | | | | RIGHT |
| | " | SPACE | TAB | DEL | CAPS | SELECT | PRINT | | ARROW |

MSX KEYBOARD MATRIX

| MEMORY LOCATION | | VALUE CONTAINED IN MEMORY LOCATION | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 254 | 253 | 251 | 247 | 239 | 223 | 191 | 127 |
| M | 64474 | ) | ! | @ | # | $ | % | ^ | & |
| E | " | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| M | 64475 | * | ( | _ | + | ; | { | } | : |
| O | " | 9 | 9 | - | = | \ | [ | ] | : |
| R | 64476 | " | ~ | ^ | > | ? | DEAD | | |
| Y | " | , | < | . | . | / | KEY | A | B |
| | 64477 | C | D | E | F | G | H | I | J |
| L | 64478 | K | L | M | N | O | P | Q | R |
| O | 64479 | S | T | U | V | W | X | Y | Z |
| C | 64480 | | | | | | F6 | F7 | F8 |
| A | " | SHIFT | CTRL | GRAPH | CAPS | CODE | F1 | F2 | F3 |
| T | 64481 | F9 | F10 | | | | BACK | | |
| I | " | F4 | F5 | ESC | TAB | STOP | SPACE | SELECT | RETURN |
| O | 64882 | | | | | LEFT | UP | DOWN | RIGHT |
| N | " | SPACE | CLS | INS | DEL | ARROW | ARROW | ARROW | ARROW |

For example, to check for the A key being pressed on an SV318/328 use -

```
10 PRINT PEEK(64887) : GOTO 10
```

which will display 253 if the key is pressed. For an MSX computer use -

```
10 PRINT PEEK(64476) : GOTO 10
```

which will display 191 when the A key is pressed.

A further illustration of how PEEK works is to examine the area of memory which would contain the one line program

```
10 REM LINE 10
```

The area of memory used to store programs starts at location 32768 for an MSX or SV328 computer and location 49152 for an SV318.

Type NEW and enter the one line program.

```
For MSX and SV328 type FOR I=32768 TO 32784:PRINTI,PEEK(I):NEXT
For SV318 type         FOR I=49152 TO 49168:PRINTI,PEEK(I):NEXT
```

| SV318 | MSX & SV328 | | | |
|---|---|---|---|---|
| 49152 | 32768 | 0 | | start of Basic program |
| 49153 | 32769 | 15 | ) | Two byte integer indicating position of |
| 49154 | 32770 | 128 | ) | next line number ie. 15+128*256=32783 |
| 49155 | 32771 | 10 | ) | Two byte integer indicating line |
| 49156 | 32772 | 0 | ) | number ie. 10+0*256=10 |
| 49157 | 32773 | 143 | REM | Token used for REM is 143 |
| 49158 | 32774 | 32 | space | ) |
| 49159 | 32775 | 76 | L | ) ASCII values |
| 49160 | 32776 | 73 | I | ) |
| 49161 | 32777 | 78 | N | ) for these |
| 49162 | 32778 | 69 | E | ) |
| 49163 | 32779 | 32 | space | ) characters |
| 49164 | 32780 | 49 | 1 | ) |
| 49165 | 32781 | 48 | 0 | ) |
| 49166 | 32782 | 0 | ) | End of program line |
| 49167 | 32783 | 0 | ) | Two zeroes for |
| 49168 | 32784 | 0 | ) | end of program |

This shows how a Basic program is stored in memory. Leave this program in memory for use with the next command explanation.

## The POKE command

Before discussing the POKE command, be warned that POKEing locations without knowing what effect it will have CAN BE DISASTROUS if you have a program in memory or if you are operating in Disk Basic. If you POKE the wrong location, it can delete the program, lock up the computer or crash your disks. So be careful.

This command is used to change the contents of a memory location by POKEing information into it. It takes the form - POKE memory location, information where memory location can be given as a value between 0 and 65535 and information is a numeric value between 0 and 255 as in POKE 33000,127 or a numeric variable which has a value between 0 and 255 as in POKE 33000,I.

With the previous one line program still in memory, let us alter the program without manually editing it.

For MSX and SV328 type POKE 32773,137 and press enter. For SV318 machines type POKE 49157,137 and press enter.

Now LIST the program. It will have changed to -

```
10 GOTO LINE 10
```

because 137 is the TOKEN for the command GOTO

Another use for the POKE command is to change the value of 'flags' in the System RAM which the computer uses to keep track of what it should be doing.

Spectravideo memory location 65079 and MSX location 64682 contains a 'flag' to show which cursor style is to be displayed - full height or half height. A zero indicates full height or normal cursor. A one indicates half height or 'insert' cursor style.

Try this

Spectravideo

```
10 CLS
20 POKE 65079,0
30 LOCATE 0,0
40 POKE 65079,1
50 GOTO 20
```

MSX

```
10 CLS
20 POKE 64682,0
30 LOCATE 0,0
40 POKE 64682,1
50 GOTO 20
```

## The VPEEK & VPOKE commands

These commands are similar to the PEEK and POKE commands except that they are used with the Video memory or VRAM.

Using VPEEK & VPOKE in text mode

```
10 REM ensure caps lock is OFF
20 CLS : WIDTH 40
30 REM ENTER SOME LINES OF TEXT
40 INPUT X$
50 REM READ AND CHANGE THE FIRST SECTION OF THE VRAM
60 FOR L=2TO500
70 VPOKE L,VPEEK(L)-32
80 NEXT L
90 GOTO 90
```

RUN this and type in enough lower case characters to fill two screen lines before pressing ENTER. You will see this routine convert all lower case characters to upper case.

To use VPEEK and VPOKE in the Graphics mode requires a good understanding of the Video RAM memory map. These will be covered in the section on sprites.

# GRAPHICS COMMANDS

This month we continue the Graphics Command section with a further explanation of the DRAW command and introduce Sprites. The term GRAPHICS CURSOR is used here. This is different from the cursor which you are used to seeing when you type in a program. In graphics mode, the graphics cursor is not actually a cursor, but the term is used to describe the point at which A GRAPHICS COMMAND WILL BE DISPLAYED.

## The DRAW command

Last month we covered the direction instructions of the Graphics Macro Language. The other instructions are -

N = draw and come back
M = move to location
B = move without drawing
A = rotate
S = scale
C = color
X = execute macro

The N instruction can be used as follows -

```
DRAW "NR20ND20"
```

This will draw 20 pixels (dots) to the right and come back, 20 down and come back. The graphics cursor (which you cannot see) will now be positioned back at the point where it started.

The M instruction

This is the move instruction and is used to move the Graphics cursor or starting point to the co-ordinates following the M instruction. A line will be drawn in the current foreground colour from the last graphics position to this new position. If the DRAW command is the first graphics command used since power-up, then the line will be drawn from the top-left corner (0,0) to this new position.

```
10 SCREEN 1,2 : REM MSX USE SCREEN 2,2
20 PSET (0,0)
30 DRAW "M50,50D50"
```

In this example we used positive numbers for the two co-ordinates following the M instruction, meaning screen co-ordinates 50 across and 50 down from the top-left corner of the screen. A + or - sign before each of the values means to move in relation to the current position on the screen.

```
10 SCREEN 1,2 : REM MSX USE SCREEN 2,2
20 PSET (128,96)
30 DRAW "M+50,-50"
```

## The B instruction

This is the BLANK instruction which can be used to prefix other instructions to inhibit drawing. For example BM will move but not draw from the old position to the new. BD20 will move the graphics cursor down 20 pixels but a line will not appear.

```
10 SCREEN 1,2 : REM MSX USE SCREEN 2,2
20 PSET (0,0)
30 DRAW "BM50,50D50"
40 GOTO 40
```

## The A instruction

This is the ANGLE instruction, and after it is used, all direction instructions will have a different effect.

```
10 SCREEN 1 : REM FOR MSX USE SCREEN 2
20 FOR A=0 TO 3 : READ C
30 D$="A"+STR$(A)+"C"+STR$(C)
40 DRAW"BM96,96XD$;R20"
50 NEXT A
60 DATA 1,13,14,15
70 GOTO 70
```

This will draw a cross with a different colour for each leg. The ANGLE instruction has actually changed the direction of drawing.

## The S instruction

This is the SCALE instruction which decides how many pixels (or dots) are turned on when DRAWing in any direction. The normal scale is S4 and will turn on one pixel times the value following a direction instruction. For example - DRAW "R20" will turn on twenty pixels to the right, DRAW "S2R20" will draw 10 pixels to the right and DRAW "S8R20" will draw forty pixels to the right.

```
10 SCREEN 1 : REM FOR MSX USE SCREEN 2
20 DRAW"S1" : GOSUB 90
30 DRAW"S2" : GOSUB 90
40 DRAW"S3" : GOSUB 90
50 DRAW"S4" : GOSUB 90
60 DRAW"S25" : GOSUB 90
70 DRAW"S4"
80 END
90 CLS : DRAW"BM20,20R10D10L10U10"
100 FOR T=1 TO 100 : NEXT T
110 RETURN
```

This will draw boxes of different sizes using just the one set of directions given in line 90.

The C instruction

This instruction is used to change the foreground colour within a DRAW string expression. For example

```
100 DRAW"BM23,20C1R24C13R24C14R24C15R24"
```

will draw a line divided into four different colours.

The X instruction

This instruction is used to execute draw instructions which are contained in a previously defined string. The string variable must contain valid draw commands and is executed by prefixing it with X and following it with a semi-colon.

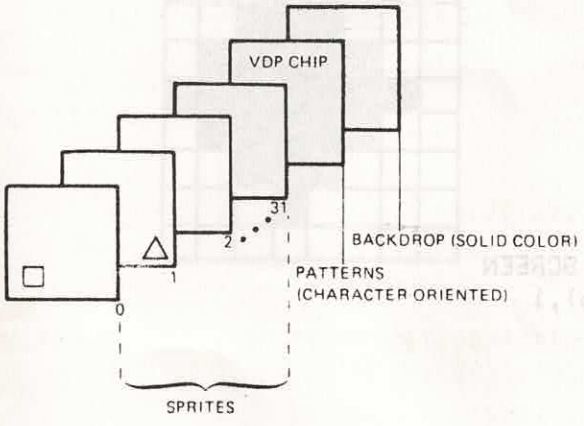
```
10 SCREEN 1 : REM FOR MSX USE SCREEN 2
20 S$="R20D20"
30 T$="R45U10"
40 DRAW"XS$;XT$;"
50 GOTO 50
```

The strings S$ and T$ will be drawn just as if the instructions contained in those strings had been within the DRAW command string.

## SPRITES

The way in which Sprites can be used is what makes the Spectravideo and MSX computers vastly superior in performance than other graphic systems. Sprites are blocks of pixels or dots arranged to form a pattern. These blocks can then be moved around on the screen and made to move in front of or behind each other.

In graphics mode, the Video Display Processor (VDP) divides the display into a BACKDROP which is coloured according to the background colour of the COLOR command, the PATTERN PLANE which is used by the LINE, DRAW, CIRCLE, PSET and PRESET commands, and 32 SPRITE PLANES (numbered 0 to 31) each of which can display one sprite. See diagram -



You can design up to 256 sprites (8x8) or 64 sprites (16x16) but only 32 of these can be displayed at a time (one per sprite plane).

The first step is to design a sprite -

## The SPRITE$ command

This command is used to create the graphic blocks by turning on or off set points within a grid. That grid (or Sprite) can be 8x8 or 16x16 pixels.

The simplest Sprite to design is an 8x8 square block in which all pixels are turned on. One way to do this is -

```
10 REM SETUP SCREEN FOR GRAPHICS
20 SCREEN 1,2 : REM FOR MSX USE SCREEN 2,2
30 REM BUILD THE BOX SPRITE
40 SPRITE$(1)=STRING$(8,255)
50 REM PLACE IT ON THE SCREEN
60 PUT SPRITE 1,(128,96),1
70 REM KEEP IT THERE
80 GOTO 80
```

This displays a box sprite in the centre of the screen.

Usually we want sprites to be a bit more interesting than just box shapes. To do this we need a clear idea of what the sprite will look like.

If we build an 8x8 sprite using data statements which contain binary numbers, then this is made easier although it is not the neatest of programming styles. To design a diamond shaped sprite with this technique -

```
10 REM SETUP SCREEN FOR GRAPHICS
20 SCREEN 1,2 : REM FOR MSX USE SCREEN 2,2
30 REM BUILD THE SPRITE
40 FOR I=1 TO 8
50 READ B$ : B=VAL("&B"+B$)
60 S$=S$+CHR$(B)
70 NEXT I
80 SPRITE$(1)=S$
90 REM DATA STATEMENTS FOR SPRITE
100 DATA 00010000
110 DATA 00111000
120 DATA 01111100
130 DATA 11111110
140 DATA 01111100
150 DATA 00111000
160 DATA 00010000
170 DATA 00000000
180 REM PLACE IT ON THE SCREEN
190 PUT SPRITE 1,(128,96),1
200 REM KEEP IT THERE
210 GOTO 210
```

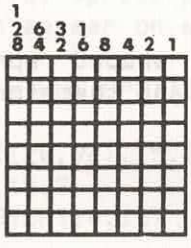
It is quite easy to see why this is the easiest way to design 8x8 sprites. You can see the pattern emerging within the data statements. This is not so easy when it comes to designing 16x16 sprites.

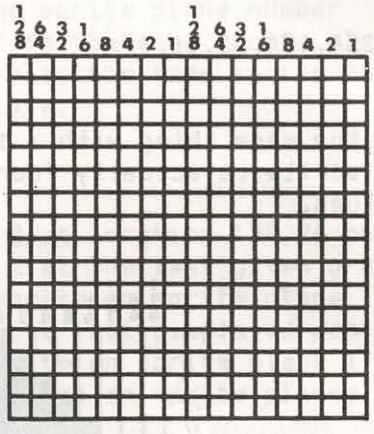
The 16x16 sprite is actually four 8x8 sprites arranged as follows

| | |
|---|---|
| Section 1 | Section 3 |
| Section 2 | Section 4 |

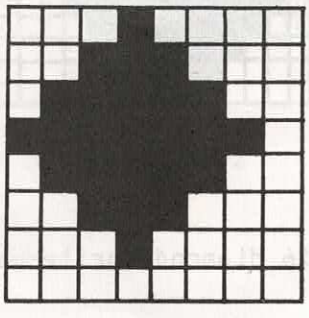
To design a 16x16 diamond shaped sprite using the previous method would require thirty two lines of data statements. Even if we did this we would still not get a clear picture of the sprite because the data statements contain information for the left side of the sprite followed by information for the right side.

A better system for designing sprites is to use a sprite grid to arrive at decimal values for the data statements -

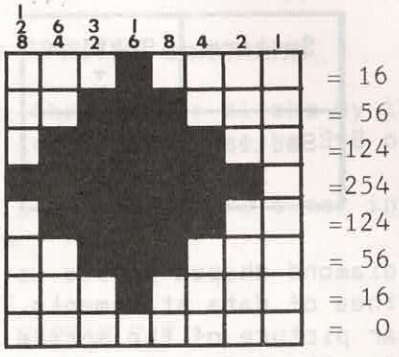



Let us use this to design the diamond sprite -



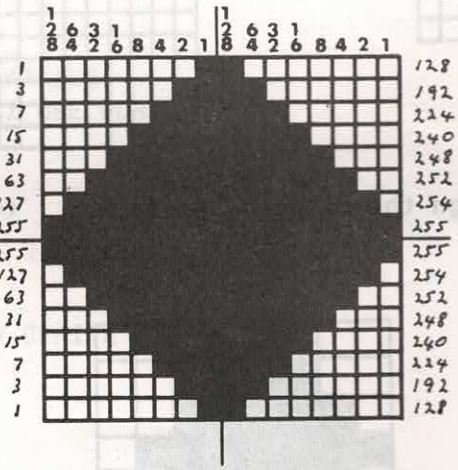
Taking each row in turn, move across and add together the values at the top of each column where the square has been filled in. The result will look like this -




Therefore, the DATA statement for the decimal values would look like -

```
DATA 16,56,124,254,124,56,16,0
```

Now we can try the same thing with a 16x16 diamond shaped sprite. Remember that a 16x16 sprite is actually four 8x8 sprites and therefore we need 32 decimal data values.




The DATA statements for the 16x16 diamond sprite would be -

```
DATA 1,3,7,15,31,63,127,255
DATA 255,127,63,31,15,7,3,1
DATA 128,192,224,240,248,252,254,255
DATA 255,254,252,248,240,224,192,128
```

## The PUT SPRITE command

This command is used to place the sprite on the screen. It takes the form -

```
PUT SPRITE a,(b,c),d,e
```

where a = sprite plane number
b = position across the screen
c = position down the screen
d = colour to use for sprite
e = pattern number to use for sprite

The sprite plane number is used to signify the sprite priority. That is, a sprite with a lower sprite plane number has priority over those with a higher plane number and, if occupying the same position on the screen, will block out the sprites with higher plane numbers.

The position on the screen can be outside the range 0-255 across or 0-191 down, but in that case part or all of the sprite will be off the screen. The pattern number is the same as that used to define the sprite with the SPRITE$ command.

Many people get confused between the sprite plane number and the sprite pattern number. The point to remember is that the sprite plane number comes first and several sprite planes can display the same sprite pattern, but only one sprite can appear on each sprite plane.

Examples of the PUT SPRITE command -

```
PUT SPRITE 1,(128,96),15,2 puts sprite 2 at location 128,96 on sprite plane 1
PUT SPRITE 1,,15,2         puts sprite 2 at the last given graphics position
PUT SPRITE 1,STEP(10,10)   places the sprite on sprite plane 1, 10 down and 10
                           across from its last displayed position
PUT SPRITE 1,(128,209)     hides the sprite on sprite plane 1
PUT SPRITE 1,(128,208)     hides the sprites on sprite planes 1 to 31
```

## MOVING SPRITES

Now that we can design sprites, we can see about moving them. The last example program was not very exciting, as we could have done the same thing with the LINE or DRAW commands. The advantage of using sprites is the ability to MOVE them around. This, is done with the PUT SPRITE command.

```
10 REM SETUP SCREEN FOR GRAPHICS
20 SCREEN 1,2 : REM FOR MSX USE SCREEN 2,2
30 REM BUILD THE BOX SPRITE
40 SPRITE$(1)=STRING$(8,255)
50 REM PLACE IT ON THE SCREEN
60 PUT SPRITE 1,(128,96),1
70 REM MOVE IT
80 FOR M=128 TO 200
90 PUTSPRITE1,(M,96),1
100 NEXT M
```

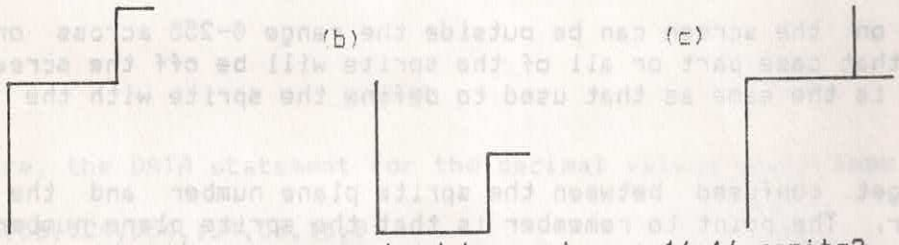
QUESTIONS

1. Can PEEK be used to examine the ROM area addresses 0 - 32767 ?

2. Why must the POKE command be used with caution?

3. Can PEEK and POKE be used to examine the Video RAM?

4. DRAW"ND50R30NU20R10" will draw what?

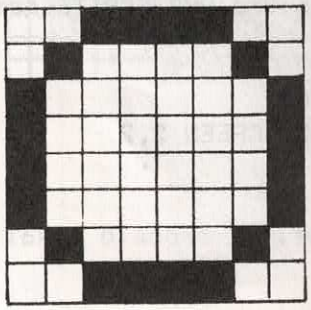



5. How many data items are required to produce a 16x16 sprite?

6. PUT SPRITE 1,(230,170),15,1 will place the sprite where?

   (a) Middle of screen
   (b) Lower right section
   (c) Upper right section
   (d) Upper left section
   (e) Lower left section

7. What would be the decimal data values for this sprite?



# MAILBOX



## INSTRUCTIONS WANTED

I recently acquired games for my Spectravideo and did not acquire instructions. The games are — CANDOO NINJA and OLD MAC FARMER. I would be grateful if you could send me a copy of these games' instructions or publish my letter in your magazine in case someone can help.

Jeremy Quan Sing
Maida Vale, W.A.

*Can anyone help?*

## CONGRATULATIONS

Congratulations are in order for your great magazine. Just as I thought I was going down for the last time, along comes the FORUM. My interest in the SV has been renewed and my typing finger is now very tired. I am looking forward to your article on the memory expansion for my SV318. Anything in the pipeline for an RS232 project or circuit? Keep up the good work.

Greg Coe
Victoria

*Plans for an RS232 project are in the pipeline, Greg. We are delaying publication, pending the feasibility of providing a short form kit for the project.*

**END**

# HINTS & TIPS

## FREE — EXTRA MEMORY FOR MSX DISK SYSTEMS

With MSX disk systems, the disk system can stop you loading a large tape program. One way of overcoming this is to hold down the SHIFT key on power-up until the beep is heard.

On the SV 738, this alters the sign-on message from 24456 Bytes Free to 28815 Bytes Free. What happens if you still want to use disks but need some more bytes of memory?

Well, it seems that the MSX system caters for multiple drive usage with the resultant loss of some memory space. Holding down the CTRL key on power-up will still give you disk capability, but add almost 1K of available memory. On the 738 this results in 25502 Bytes Free.

## AND FOR SPECTRAVIDEO

When your system boots to disk, you get a message advising something like 21191 Bytes Free. Typing FRE(0) will also confirm this fact. If you now type MAXFILES = 0 and press enter, the FRE command will now return 267 more than previously.

This is because the disk system pre-supposes that you wish to read and write files and, in essence, performs a MAXFILES = 1 function. If the program you are using does not need this facility, then you can gain extra memory by killing this file buffer. In fact, try doing several MAXFILES with different values and checking the resultant RAM value with FRE(0).

## ORGANISE YOUR DISKS

Ever been caught after having typed in a program only to be hit with the message Disk Full when you try to Save it? It is a good idea, when you purchase some new disks, to format the lot ready for use. This has two advantages. If the disk won't format properly, you can take it back to the dealer. You also have a set of formatted disks ready for that dreadful Disk Full message.

## SPECTRAVIDEO LOGO

I would like to give you a tip on how to get the Spectravideo Logo on the screen. It is as follows —

DEFUSR = &H4782:PRINT USR(&H4782):SCREEN 0

S. Dimitrio
Footscray, Victoria

## JUST A QUICK ONE ...

Don't use paper clips anywhere near your cassettes or disks. These may have been stored in those containers with a magnet around the top. You have probably seen them at stationers — the idea being to shake up the container and some clips cling to the magnet for easy removal. I have even seen word processor operators shake out a paper clip and use it to attach a note to their word processing disks and then complain about lost information.

**END**

# HARDWARE MODS

## MORE MEMORY FOR YOUR 318 MKII

*John Tillotson describes how to add more memory to your SV318 MKII.*

I have an upgraded 318 machine and having seen how easily it was done wondered if your readers might like to upgrade their machines themselves as it is not a difficult task.

Firstly, carefully de-solder the eight 4116 RAM chips and clear the mounting holes out completely. My new 4164 chips were mounted in blocks and I recommend this as it saves heat sink problems.

Finally, there are three jumper wires to remove and replace. Carefully inspect the board to find these jumpers notated "16K". On the MKII they are quite clearly marked as are the replacement spots marked "64K".

If one is careful in removing the 4116 chips then they could be replaced in the blocks and the machine returned to 318 specifications.

## UPGRADE YOUR 318 (MKI)

### Go to 64K

*John's article spurred us to look at the 318 MKI and see what steps were required to go to 64K.*

This article describes how to upgrade the SV318 MKI memory to the same standard as the SV328 using either 4164 or 4864 memory chips (TMM type). We warn that although the procedure described is straightforward, no responsibility is accepted for damage to the computer or failure of the system to work. If you are not experienced with a soldering iron and can't comfortably fiddle inside your computer then — DON'T DO IT!

The steps involved are —

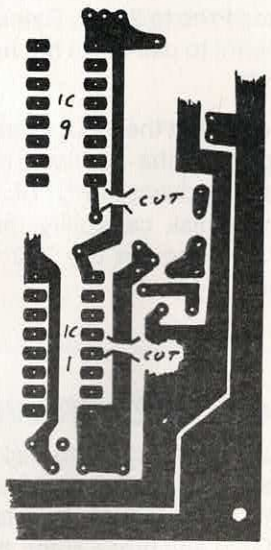
1. Remove the old 4116 memory chips.
2. Remove some capacitors.
3. Re-arrange power supply for the memory chips.
4. Activate the A14/A15 address select signal.
5. Activate the BANK21 (low) signal.
6. Modify the RFRSH (low) signal.
7. Install the new memory chips.

Opening the case will not be described because if you can't figure that out, then you shouldn't be doing this modification. Suffice to say that when you have opened the case, take care not to strain the cable connecting the keyboard to the main PCB.

1. Remove the 4116 memory chips marked ICI — IC8 only and clean out all holes. DO NOT REMOVE IC9-16 which are the VRAM.
2. Remove capacitors C53-C60 which are not needed in the upgraded system.
3. The 4116 chips used +12V, +5V and -5V at pins 8, 9 and 1 respectively. The new 64K chips only require +5V at pin 8.

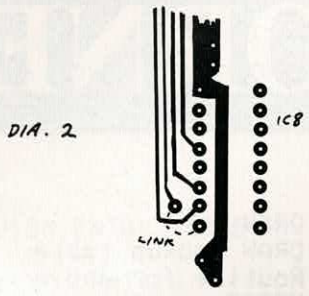
   From the component side of the board, cut the track to the left of where C53 was removed to disconnect all pin 9 from the +5V rail. To supply +5V to pin 8 of the new chips, link the track at the left of C61 to the pad on the left of the cut just made. Check that C53 IS NOT connected to C20 and that C61 IS.

   Turn the board over with the joystick port on your left and the expansion port to the top away from you. Refer to Diagram 1 and locate this area at the bottom right corner of the board. Double check that you have the correct area and cut the two tracks as shown. The lower cut disconnects pin 8 from the +12V supply and the top cut disconnects pin 1 from the -5V supply. To complete this separation, move to your left and locate IC16. Cut the track which connects IC8 pin 1 to IC16 pin 1. Do this next to pin 6 of IC16.

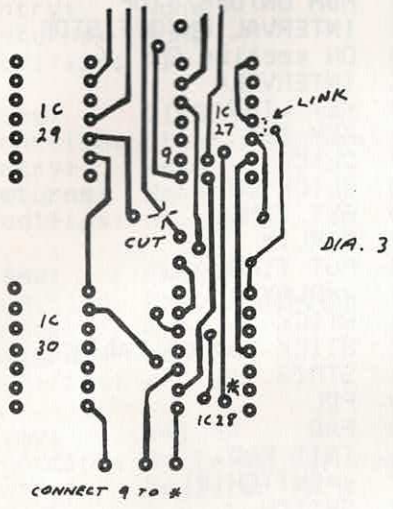



DIA. 1

4. Move down to IC8 and link pin 9 to the hole near pin 10 as shown in Diagram 2. This will connect pin 9 of ICI to IC8 with pin 12 of IC17. This is the A14/A15 multiplexer.




5. Move to the left. Locate IC27 and IC28 (diagram 3). Link pins 4 and 5 of IC27 to activate the BANK21 (low) signal.




6. Run an insulated wire from IC27, pin 9 to the hole next to pin 7 of IC28. Locate the hole approximately 9mm below pin 8 of IC27. Cut the track coming away from this hole to sever IC27 pin 5 from IC47 pin 8, to allow RFRSH to function properly.
7. Read through these instructions again and follow your work on the PCB. If all is well — you are ready to install your new 64K chips. These can either be soldered in place or solder in sockets and plug in the chips. Ensure that the chips are oriented the same as IC9 — IC16, with the notch at the top.

   You should now be able to power up and see a very much larger BYTES FREE message displayed.

## NEW KEYBOARD

If you are happy with what you have just done, then you might care to go the whole way and fit the SV328 type keyboard. This is simplicity itself. First buy a 328 replacement keyboard — part number KB328. Open up the 318 and remove the ten screws which hold the keyboard in the top of the case. Use these screws to secure the new keyboard in place. Cut the cable connecting the old keyboard to the main PCB.

Desolder the remains of this cable from the main PCB, and solder in the cable which is attached to the new keyboard. Reassemble the case.

Now, the only difference between your computer and a 328 is the label on top.

# MEMORY DECODE (SV & MSX)

*Background to the memory upgrade article but decoding is similar for all SV and MSX computers.*

To address up to 64K of RAM, the computer requires sixteen address lines — A0 to A15. The maximum number which can be generated by a sixteen bit binary number is 65535 or 64K. Providing a high (1) on all lines up to A13 will access 16K, adding a 1 to A14 will access 32K, 1 on A15 and 0 on A14 will access 48K and 1 on all lines accesses 64K.

| | A15 | A14 | A13 — AO |
|---|---|---|---|
| up to 16K (16383) | 0 | 0 | 0 or 1 |
| up to 32K (32767) | 0 | 1 | 0 or 1 |
| up to 48K (49151) | 1 | 0 | 0 or 1 |
| up to 64K (65535) | 1 | 1 | 0 or 1 |

The memory chips can be thought of as similar to a two dimension array in Basic —

```
                COLUMN
          0 1 2 3 4 5 ... 253 254 255
        0
R       1                    ARRAY OF
O       .                    MEMORY
W     254                    LOCATIONS
      255
```

Three signals affect memory access — RAS (Row Address Strobe), CAS (Column Address Strobe) and MPX (Multiplexer). If a MREQ (Memory Request) signal is sent by the CPU while MPX is low, the CAS signal strobes the Column address into the memory chips. While MREQ is still low, MPX goes high and the Row address is strobed to the chips.

If, at this time, the RD (Read) signal is high, then any data appearing on the data lines D0 ← D7 will be placed in that memory location. If RD is low the value stored at that memory location is placed on the data lines.

There are eight memory chips — one for each data line or BIT. To illustrate this, POKEing the value 143 into memory address 39000 involves CAS strobing in 152 (39000/256) and RAS strobing 88 i.e. 39000-(152*256). The RD line carries a 1 (high) and the data lines transfer the binary equivalent of 143 into memory location 39000. The result would look like —

8 only 64K BIT CHIPS

| Memory location 39000 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|
| | DB7 | DB6 | DB5 | DB4 | DB3 | DB2 | DB1 | DB0 |

**END**

# HACKERS CORNER

*Hello once again, I'm Robert Brinkworth and I thought you might like to know about me. I live in Morley, a suburb of Perth in Western Australia and currently in my third year of an Electrical Installing apprenticeship at Midland Brick in Middle Swan, Perth.*

My personal computer equipment consists of an SV328 two-disk drive system with Epson LX80 printer and an SV/MSX 728 computer. My hobbies include listening to Heavy Metal, computing, electronics, girls and going out. Oh, and looking at SLR5000's.

Due to unforseeable delays, no machine language tutorial will appear in this issue. But next issue it will be bigger and brighter than ever.

## SUBROUTINE SECTION

I wish to start a subroutine section in which users who have found, or developed, a useful subroutine might like to contribute to this section.

## GAME LOAD ADDRESSES

If you have any start, end and execution address written down for any games, send the details in, as I would like to include them as a separate section.

## THE DOS

To remain a viable proposition as a program I need more response about the DOS as a whole and in the internal design.

## LIST LOCKOUT (SV and MSX)

If you want to protect a program you have written from being LISTed by the user, then only three POKEs are necessary.

For MSX —
10 POKE &HFF89,&HC3:POKE &HFF8A,&H9B:POKE &HFF8B,&H40

For SV —
10 POKE &HFEF4,&HC3:POKE &HFEF5, &HAF:POKE &HFEF6,&H09

This operation redirects the LIST command to jump to the BASIC Ready entry. After this line is executed and LIST is typed, "Ok" is displayed and you are back in the command line interpreter.

## SV ROM DECODED (Part Three)

```
Spectravideo SV-318/328 ROM version 1.1


24FC  PAINT (X,Y),C
254B  Routine for above
2652  CIRCLE (X,Y),S,C,start,end,H/W
273E  Routine for above
29DA  DRAW
29E7  DRAW lookup table
2A15  Routine for above
2BE5  MOTOR
2BFD  SOUND
2C24  PLAY
2D87  PLAY lookup table
2FB1  PUT
2FC6  PUT STEP
2FD1  LOCATE
3036  MDM ON/OFF/STOP
3042  INTERVAL ON/OFF/STOP
30BB  ON section 2
30EC  INTERVAL=
3120  KEY LIST
3136  KEY SET
31AF  CLICK
31B8  CLICK ON
31BD  GET TIME
31C7  CSRLIN
31D3  PUT TIME
31DE  x=PLAY(x)
3206  STICK
3253  STICK lookup table
3263  STRIG
3280  PDL
32BD  PAD
32D6  INIT PAD
3377  x=SWITCH(0)
337F  SWITCH
33EB  BANKING data
3420  Routine for above
346A  OUT(8CH),B:OUT(8CH),C
3476  BANKJP
3480  BANK CALL
3498  KEY label restore
34A6  DSKO$
34AB  SET
34B0  NAME
34B5  KILL
34BA  IPL
34BF  COPY
34C4  CMD
34C9  DSKF(x)
34CE  DSKI$
34D3  ATTR$
34D9  Setup PIO,PIA,LPT,80 COL.
3502  80 column data
3512  BREAKXX or STATUS
3536  SET video register
3541  SCREEN0 or TEXT
3610  SCREEN1 or HIRES
3665  SCREEN2 or MULTI
36BE  Setup Sprite Attribute
      Table and sprites


***** End of instalment 3 *****
```

## MSX ROM CALLS

All MSX ROM calls should be made through the BIOS entry table.

MSX1 BIOS ENTRY TABLE

```
003BH  Name:       INITIO
       Function:   Device initialisation
       Entry:      None
       Returns:    None
       Modifies:   All

003EH  Name:       INIFNK
       Function:   Init. Function Keys
       Entry:      None
       Returns:    None
       Modifies:   All

0041H  Name:       DISSCR
       Function:   Disable screen
       Entry:      None
       Returns:    None
       Modifies:   AF, BC

0044H  Name:       ENASCR
       Function:   Enable screen
       Entry:      None
       Returns:    None
       Modifies:   AF, BC

0047H  Name:       WRTVDP
       Function:   Write to VDP register
       Entry:      Register in C
                   Data in B
       Returns:    None
       Modifies:   AF, BC

004AH  Name:       RDVRM
       Function:   Read VRAM
                   addressed by HL
       Entry:      HL
       Returns:    A
       Modifies:   AF

004DH  Name:       WRTVRM
       Function:   Write to VRAM
                   addressed by HL
       Entry:      HL,A
       Returns:    None
       Modifies:   AF

0050H  Name:       SETRD
       Function:   Set up VDP for read
       Entry:      HL
       Returns:    None
       Modifies:   AF

0053H  Name:       SETWRT
       Function:   Set up VDP for write
       Entry:      HL
       Returns:    None
       Modifies:   AF

0056H  Name:       FILVRM
       Function:   Fills VRAM with data
       Entry:      Address in HL
                   Length in BC
                   Data in A
       Returns:    None
       Modifies:   AF,BC

0059H  Name:       LDIRMV
       Function:   Move block of VRAM
                   to main memory
       Entry:      Source in HL
                   Destination in DE
                   Length in BC
       Returns:    None
       Modifies:   All
```

**END**

# USER GROUPS

## SYDNEY USERS

We have been advised of a group in Sydney for Spectravideo owners. The group has members with software and hardware experience and holds workshops on specific problems and areas of interest. The aim of the group is to bring together SV owners and maintain the personal contact which has been lost since that machine now has no dealer in the Sydney area.

Meetings are generally informal and if you would like more information please contact Malcolm Samuels on 888 7664 after hours.

## NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
P.O. Box 26-050
Newlands, Wellington 6004, New Zealand

## QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker (Secretary)
25 Primrose Street, Woodridge, Queensland 4114

SUNSHINE COAST SPECTRAVIDEO & MSX USER GROUP
C/- Keith Cunnington (Secretary)
39 Waterfall Road, Nambour, Queensland 4560

## VICTORIA

MELBOURNE SPECTRAVIDEO & MSX USERS GROUP
C/- John Halford, 48 Shawlands Avenue
Blackburn South, Victoria 3130
Phone (03) 878 6169

GIPPSLAND MSX USER GROUP
C/- George Francis, 31 Donald Street
Morwell, Victoria 3840
Phone 051-34 5175

## WESTERN AUSTRALIA

THE MSX USERS GROUP
C/- Max Cowin, P.O. Box 87
Mt Hawthorn, W.A. 6016

# MORE COLOUR

NICHOLAS KARADIMAS has sent in these useful routines for use with the SV318/328. Using them will produce extra colours – about 120 in total. The first routine is the machine code loader which must be loaded and RUN before loading the other routines.

```
1 REM MACHINE CODE LOADING PROGRAM FOR
 SHADING ROUTINE.TYPE THIS PROGRAM IN AN
D CSAVE IT.THEN WHENEVER YOU WANT TO USE
 THE SHADING SUBROUTINE RUN IT AND THERE
AFTER RUN WHATEVER PROGRAM YOU HAVE
10 CLEAR 250,&HF400
20 DEF USR1=&HF450
30 FOR X=&HF450 TO &HF498
40 READ A$
50 POKE X,VAL("&H"+A$)
60 NEXT X
70 REM POKE &HF440,0:POKE &HF441,0:POKE
&HF442,1
100 DATA 3E,0,D3,81,3A,41,F4,F6,40,D3,81
,21,0,0,3A,42,F4,67,3E,AA,BA,28,4
110 DATA 16,AA,18,2,16,55,7A,D3,80,23,3E
,18,BC,20,EC,3E,0,D3,81,3A,41,F4,C6,20,F
6,40,D3,81,21,0
120 DATA 0,3A,40,F4,57,3A,42,F4,67,7A,D3
,80,23,3E,18,BC,20,F7,C9,CC
```

## DEMO 1

```
100 N=RND(-TIME):COLOR 12,1,1:SCREEN 1
101 FOR X=1 TO 3
102 W=INT((RND(1)*224)+32)
103 Y=( (X*8)MOD 24):Z=((X*8+7)MOD 24)+1
104 A=W:B=Y:C=Z:GOSUB 10000
106 NEXT X
110 GOTO 101
10000 IF (A<0)OR(A>255)OR(B<0)OR(C<1)OR(
B>23)OR(C>24) THEN RETURN
10010 ' SHADING SUBROUTINE
10011 ' A HOLDS COLOR 0-255
10012 ' B HOLDS STARTING POINT 0-23
10013 ' C HOLDS ENDING POINT 1-24
10014 ' THIS ROUTINE WILL SHADE A
10015 ' PORTION OF THE SCREEN IN
10016 ' ONE OF 256 COLOURS
10017 ' THE PORTION BEING THE
10018 ' THE HORIZONTAL SEGMENTS OF THE
10019 ' SCREEN  DIVIDED INTO 24
10020 POKE &HF440,A
10030 POKE &HF441,B
10040 POKE &HF442,24-C+B
10050 Z=USR1(0)
11000 RETURN
```

## DEMO 2

```
10 SCREEN 1
20 A=3*16+15:B=3:C=4:GOSUB 1000
30 A=2*16+15:B=4:C=5:GOSUB 1000
40 A=12*16+15:B=5:C=6:GOSUB 1000
50 A=3*16+14:B=6:C=7:GOSUB 1000
60 A=2*16+14:B=7:C=8:GOSUB 1000
70 A=12*16+14:B=8:C=9:GOSUB 1000
80 A=3*16+7:B=0:C=1:GOSUB 1000
90 A=2*16+7:B=1:C=2:GOSUB 1000
100 A=12*16+7:B=2:C=3:GOSUB 1000
110 A=3*16+11:B=9:C=10:GOSUB 1000
```

# READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

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```
120 A=2*16+11:B=10:C=11:GOSUB 1000
130 A=3*16+10:B=11:C=12:GOSUB 1000
140 A=2*16+10:B=12:C=13:GOSUB 1000
150 A=12*16+11:B=13:C=14:GOSUB 1000
160 A=12*16+10:B=14:C=15:GOSUB 1000
170 A=3*16+3:B=15:C=16:GOSUB 1000
180 A=3*16+2:B=16:C=17:GOSUB 1000
190 A=2*16+2:B=17:C=18:GOSUB 1000
200 A=3*16+12:B=18:C=19:GOSUB 1000
210 A=2*16+12:B=19:C=20:GOSUB 1000
220 A=12*16+12:B=20:C=21:GOSUB 1000
230 A=3*16+1:B=21:C=22:GOSUB 1000
240 A=2*16+1:B=22:C=23:GOSUB 1000
250 A=12*16+1:B=23:C=24:GOSUB 1000
500 GOTO 500
1000 ' sub
1010 POKE &HF440,A
1011 POKE &HF441,B
1012 POKE &HF442,24-C+B
1015 Z=USR1(0)
2000 RETURN
```

## DEMO 3

```
499 N=RND(-TIME):COLOR 12,1,1
500 SCREEN 1
510 A=7*16+14:B=0:C=6:GOSUB 1000
560 A=4*16+10:B=6:C=10:GOSUB 1000
569 A=11*16+15:B=10:C=11:GOSUB1000
570 A=11*16+7:B=11:C=12:GOSUB1000
571 A=5*16+11:B=12:C=15:GOSUB 1000
590 A=5*16+15:B=15:C=20:GOSUB 1000
591 A=10*16+14:B=20:C=21:GOSUB 1000
592 A=3*16+8:B=21:C=22:GOSUB 1000
593 A=11*16+14:B=19:C=20:GOSUB 1000
594 FOR X=0 TO 5*8
595 IF RND(1)*X<10 THEN LINE(0,X)-(256,X
),15
596 NEXT X
600 LINE(0,11*8)-(256,11*8),1
601 LINE(0,14*8)-(256,14*8),7
602 LINE(0,13*8)-(256,13*8),7
603 LINE(0,16*8)-(256,16*8),7
604 LINE(0,18*8)-(256,18*8),7
610 LINE(0,11*8+5)-(256,11*8+1),15
611 LINE(0,12*8+1)-(256,12*8+5),15
612 LINE(0,13*8+5)-(256,13*8+1),15
613 LINE(0,16*8+1)-(256,16*8+5),15
614 LINE(0,18*8+5)-(256,18*8+1),15
700 FOR X=1 TO 50
710 A=INT(RND(1)*256)
720 LINE(A,6*8 )-(A,10*8),1
730 NEXT X
740 FOR X=1 TO 50
750 A=INT(RND(1)*256)
760 LINE(A,6*8 )-(A,10*8),6
770 NEXT X
800 FOR X=0 TO 31
810 A=(INT(RND(1)*100)MOD 2)
820 B=X*8+8192+6*256
830 IF A=1 THEN VPOKE B,7*16+14
840 NEXT X
850 FOR X=0 TO 255
860 IF RND(1)>.3 THEN A=RND(1)*5:B=RND(1
)*5:C=RND(1)*6+7:LINE(X+A,175-C)-(X+B,17
5),3
861 NEXT X
900 GOTO 900
1000 ' sub
1010 POKE &HF440,A
1011 POKE &HF441,B
1012 POKE &HF442,24-C+B
1015 Z=USR1(0)
2000 RETURN
```

### PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

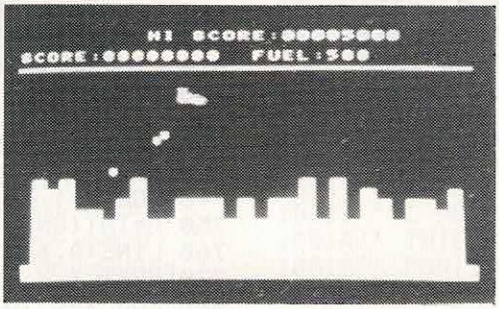
All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter I for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

We have received phone calls from some people who seem to have a problem typing in some of the listings. DO NOT waste your time and ours. If the program does not work, check your typing again and, if necessary, type the line again. WE CANNOT DE-BUG PROGRAMS OVER THE PHONE.

# CITY BOMBER




This listing is in MSX format. For Spectravideo SV318/328 owners, see the note at the end of the listing.

YOUR PLANE IS LOSING ALTITUDE AND THE ONLY PLACE TO LAND IS IN THE CITY. TO AVOID CRASHING. YOU MUST FLATTEN ALL OF THE BUILDINGS TO CREATE A LANDING SPACE. GOOD LUCK!

```
10 COLOR15,1,1:SCREEN2,2:DEFINTA-Y:DEFFN
A(X)=INT(RND(1)*X)+1:A=RND(-TIME):ZH=500
0:OPEN"GRP:"AS#1
20 A$="CITY BOMBER":COLOR9:PRESET(80,80)
:PRINT#1,A$:PRESET(81,80):PRINT#1,A$:COL
OR3:PRESET(90,180):PRINT#1,"(C) 1986"
30 RESTORE:FORA=0TO1:READA$:S$="":FORB=1
TOLEN(A$)STEP2:S$=S$+CHR$(VAL("&H"+MID$(
A$,B,2))):NEXT:SPRITE$(A)=S$:NEXT
40 CLS:A$="HI SCORE:":B$=RIGHT$(STR$(ZH)
,LEN(STR$(ZH))-1):B$=STRING$(8-LEN(B$),4
8)+B$:PRESET(80,0):COLOR13:PRINT#1,A$;:C
OLOR7:PRINT#1,B$:PRESET(81,0):COLOR13:PR
INT#1,A$;:COLOR7:PRINT#1,B$:A$="SCORE:
        FUEL:":COLOR13:PRESET(8,16):PRIN
T#1,A$
50 PRESET(9,16):PRINT#1,A$:LINE(0,28)-(2
56,30),12,BF:LINE(0,180)-(256,192),12,BF
:L=6:GOSUB170:LINE(32,144)-(224,152),1,B
F:PRESET(40,145):A$="PRESS TRIGGER TO BE
GIN":COLOR9:PRINT#1,A$:PRESET(41,145):PR
INT#1,A$
60 IFNOT(STRIG(0)+STRIG(1))THEN60ELSEL=1
:GOSUB170:ZS=0:FU=500:GOSUB180:GOSUB190:
X=0:Y=44:PUTSPRITE0,(X,Y),4:FORA=1TO3:B(
A)=0:VPOKE6912+A*4,209:NEXT
70 SOUND8,5:SOUND0,200:SOUND6,26:SOUND7,
48:A=STICK(0)+STICK(1):IFA=1ANDY>44ANDFU
>0THENY=Y-4:FU=FU-50:GOSUB190ELSEIFA=5TH
ENY=Y+4:IFY>=172THEN140
```

```
80 X=X+4:IFX>246THENX=0:Y=Y+4:IFY>=172TH
EN140
90 PUTSPRITE0,(X,Y):IF(STRIG(0)+STRIG(1)
)ANDX>8ANDX<238THENFORC=1TO3:IFB(C)=0THE
NB(C)=1:BX(C)=X+6:BY(C)=Y+14:PUTSPRITEC,
(BX(C),BY(C)),6,1ELSENEXT
100 FORC=1TO3:IFB(C)=1THENBY(C)=BY(C)+4:
IFBY(C)>186THENB(C)=0:VPOKE6912+C*4,209E
LSEVPOKE6912+C*4,BY(C):IFPOINT(BX(C)+4,B
Y(C)+4)=11THENZS=ZS+100:GOSUB180:LINE((
BX(C)+4)\8)*8,(BY(C)\8)*8)-(((BX(C)+4)\8
)*8+7,(BY(C)\8)*8+15),1,BF:B(C)=0:VPOKE6
912+C*4,209
110 NEXT:IFPOINT(X+16,Y+12)=11ORPOINT(X+
8,Y+12)=11THENSOUND8,0:SOUND9,0:SOUND0,0
:SOUND1,0:SOUND6,26:SOUND7,14:SOUND8,10:
SOUND9,20:SOUND11,130:SOUND12,150:SOUND1
3,1:FORA=1TO200:COLOR,,FNA(15):NEXT:COLO
R,,1ELSE70
120 LINE(64,80)-(200,90),1,BF:COLOR9:A$=
"G A M E  O V E R":PRESET(72,82):PRINT#1
,A$:PRESET(73,82):PRINT#1,A$:FORA=1TO200
0:NEXT:IFZS>ZHTHENZH=ZS:FORA=1TO100:SOUN
D8,8:SOUND0,FNA(255):NEXT:SOUND8,0
130 FORA=0TO5:VPOKE6912+A*4,209:NEXT:GOT
040
140 SOUND8,0:SOUND9,0:A$="CONGRADULATION
S !!!!":COLOR13:LINE(32,80)-(224,136),1,
BF:PRESET(40,88):PRINT#1,A$:PRESET(41,88
):PRINT#1,A$:A$="YOU HAVE LANDED":COLOR3
:PRESET(56,104):PRINT#1,A$:PRESET(57,104
):PRINT#1,A$:A$="BONUS":B$=STR$(FU*10*L)
:COLOR13
150 PRESET(72,120):PRINT#1,A$;:COLOR7:PR
INT#1,B$:PRESET(73,120):COLOR13:PRINT#1,
A$;:COLOR7:PRINT#1,B$:ZS=ZS+FU*10*L:GOSU
B180:L=L+1:IFL>8THENL=8
160 FORA=1TO1000:NEXT:FU=500:X=0:Y=44:PU
TSPRITE0,(X,Y),4:FORA=1TO3:B(A)=0:VPOKE6
912+A*4,209:NEXT:GOSUB170:LINE(0,180)-(2
56,192),12,BF:GOTO70
170 LINE(0,31)-(256,179),1,BF:FORA=2TO30
:LINE(A*8,179)-(A*8+7,179-(FNA(L+4)+3)*8
),11,BF:NEXT:RETURN
180 LINE(56,16)-(104,24),1,BF:COLOR7:PRE
SET(56,16):A$=RIGHT$(STR$(ZS),LEN(STR$(Z
S))-1):A$=STRING$(8-LEN(A$),48)+A$:PRINT
#1,A$:RETURN
190 LINE(176,16)-(256,24),1,BF:COLOR7:PR
ESET(176,16):A$=RIGHT$(STR$(FU),LEN(STR$
(FU))-1):PRINT#1,A$:RETURN
200 DATA FC70787C7FBF7FBF7FBF7F000000000
0000000000070C8C4FEFFFC7E
210 DATA 3C183C3C18
```

NOTE: FOR SV318/328 CHANGE LINES 10,20,40,50,120,140,180 AND 190 AS FOLLOWS -

```
10 COLOR15,1,1:SCREEN1,2:DEFINTA-Y:DEFFNA(X)=INT(RND(1)*X)+1:A=RND(-TIME):ZH=5000
```

```
20 A$="CITY BOMBER":COLOR9:LOCATE80,80:PRINTA$:LOCATE81,80:PRINTA$:COLOR3:LOCATE90,180:PRINT"(C) 1986"
```

```
40 CLS:A$="HI SCORE:":B$=RIGHT$(STR$(ZH),LEN(STR$(ZH))-1):B$=STRING$(8-LEN(B$),48)+B$:LOCATE80,0:COLOR13:PRINTA$;:COLOR7:PRINTB$:LOCATE81,0:COLOR13:PRINTA$;:COLOR7:PRINTB$:A$="SCORE:                FUEL:":COLOR13:LOCATE8,16:PRINTA$
```

```
50 LOCATE9,16:PRINTA$:LINE(0,28)-(256,30),12,BF:LINE(0,180)-(256,192),12,BF:L=6:GOSUB170:LINE(32,144)-(224,152),1,BF:LOCATE40,145:A$="PRESS TRIGGER TO BEGIN":COLOR9:PRINTA$:LOCATE41,145:PRINTA$
```

```
120 LINE(64,180)-(200,90),1,BF:COLOR9:A$="G A M E  O V E R":LOCATE72,82:PRINTA$:LOCATE73,82:PRINTA$:FORA=1TO2000:NEXT:IFZS>ZHTHENZH=ZS:FORA=1TO100:SOUND8,8:SOUND0,FNA(255):NEXT:SOUND8,0
```

```
140 SOUND8,0:SOUND9,0:A$="CONGRADULATIONS !!!!":COLOR13:LINE(32,80)-(224,136),1,BF:LOCATE40,88:PRINTA$:LOCATE41,88:PRINTA$:A$="YOU HAVE LANDED":COLOR3:LOCATE56,104:PRINTA$:LOCATE57,104:PRINTA$:A$="BONUS":B$=STR$(FU*10*L):COLOR13
```

```
180 LINE(56,16)-(104,24),1,BF:COLOR7:LOCATE56,16:A$=RIGHT$(STR$(ZS),LEN(STR$(ZS))-1):A$=STRING$(8-LEN(A$),48)+A$:PRINTA$:RETURN
```

```
190 LINE(176,16)-(256,24),1,BF:COLOR7:LOCATE176,16:A$=RIGHT$(STR$(FU),LEN(STR$(FU))-1):PRINTA$:RETURN
```

— — — — — — — — — — —

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